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Palaeomagnetic Field Directions in Siluro–Devonian Lavas of the Lorne Plateau, Scotland, and their Regional Significance

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Summary

This palaeomagnetic study of Siluro–Devonian lavas on the Lorne Plateau, Scotland, yields a pole position 2° S, 141° E ($d\psi = 5°$, $d\chi = 8°$) after magnetic cleaning of specimens from 15 sites where tectonic dip can be estimated. This conclusion is supported by a fold test at these sites, stability against AF demagnetization, the observation of similar remanence characteristics at sites where tectonic dip is not known, a contact test of variation of NRM between a Tertiary dyke and adjacent baked and unbaked lava, and mineralogical and rock magnetic considerations. All these lavas had the same polarity which, together with radiometric age determinations, suggests that this sequence is at least as old as the Crawton Group in the lower part of the ' Lower Old Red Sandstone ' age volcanic sequence in the Midland Valley of Scotland. Their direction of magnetization is concordant with published results from other proven Siluro–Devonian rocks in Great Britain.

Introduction

The Dalradian schists of the Lorne Plateau are overlain unconformably by a mainly volcanic sequence of basalts and basaltic and hypersthene andesites with intercalated ashes, silstones, shales and limestones and, locally, with a basal conglomerate. The lower part of the sequence is the more basaltic, the lavas containing phenocrysts of olivine, augite and plagioclase in a mafic groundmass. The higher flows are hypersthene andesites in which phenocrysts of plagiolcase and highly altered hypersthene and augite commonly occur, set in a predominantly glassy matrix.

The ' Lower Old Red Sandstone ' age of this series was originally inferred from fish and arthropod remains (Kynaston & Hill 1908) and subsequently confirmed by isotopic dating. The first ages were determined by Evans *et al.* (1971) at 394 ± 9 Ma by the K–Ar method. Recently a more extensive study (J. F. Brown 1974, private communication) has yielded both K–Ar and Rb–Sr isochrons which correspond to 409 ± 13 and 401 ± 5 Ma respectively (half life of Rb^{87} taken as $4{\cdot}7 \times 10^{10}$ yr). All these data are concordant and correspond to Late Silurian or Lower Devonian depending on which time scale is adopted; nevertheless the new data do suggest a slightly older age than the published estimate. The present study extends the palaeomagnetic coverage of British Siluro–Devonian rocks, and expands the limited sampling of the Lorne Plateau lavas begun by Embleton (1968).

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Results

In this study the lavas were sampled at 32 localities, using a portable drill and orienting by sun or magnetic compass and clinometer. At 15 of these localities, tectonic dip as estimated by Lee & Bailey (1925) can be allowed for. At locality 9 a Tertiary dyke was also sampled (9C) together with the lava adjacent to it (9B). All site mean NRM directions (Fig. 1(a) and Table 1) were significant at the 95 per cent confidence level (Watson 1956) with the exception of site 12, which just fails to meet this criterion. Progressive alternating field (AF) demagnetization was carried out in fields up to 1000 Oe (peak) on one or two specimens from most localities. In the majority of cases specimens whose NRM differed in direction from the others at the same site were selected for this treatment, and in general the differences were progressively reduced during demagnetization (Fig. 1(d)).

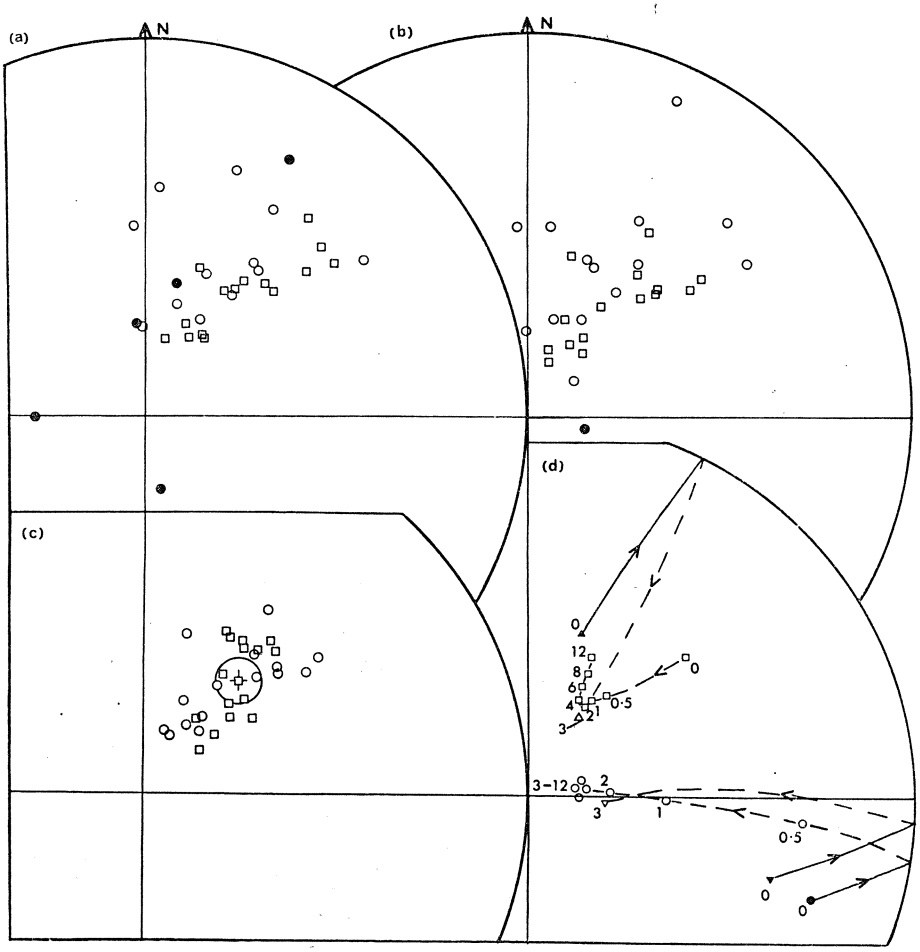


FIG. 1. Stereographic projections (open symbols upper hemisphere; solid symbols lower hemisphere). (a) Site mean directions of total NRM of lavas *in situ*; squares refer to sites where tectonic dip is known, and circles to other sites. (b) Ditto, after magnetic cleaning. (c) Mean directions after magnetic cleaning at sites where tectonic dip is known; circles denote *in situ* directions and squares denote tilt correction has been applied, from which the overall mean and circle of 95 per cent confidence are derived. (d) Progressive AF demangetization at site 12. Each specimen is given a distinct symbol; numbers denote peak alternating field in hundreds of oersteds.

Table 1

Site statistics[1] *(tilt corrected) at sites with tectonic control*

				Total NRM				AF cleaned						
Site	National Grid Reference	Strike	Dip	No. of samples	R	D,	I	Peak AF (oe)	No. of samples	R	D,	I	k	α_{95}
1	NM867316	210	20SE	5	4·97	37,	−29	300	5	4·97	40,	−35	142·3	6·4
2	NM859278	220	15SE	5	4·28	34,	−21	200	4	3·95	34,	−39	65·5	11·4
3	NM863277	220	15SE	7	5·59	45,	−27	250	5	4·97	43,	−36	155·2	6·1
4	NM869240	30	10NW	6	4·05	40,	−67	250	5	4·99	53,	−69	1319·8	2·1
5	NM865266	220	15SE	6	5·99	34,	−40	300	6	5·99	35,	−41	378·7	3·4
6	NM889339	230	20SE	7	6·93	28,	−35	300	7	6·82	27,	−39	32·5	10·7
7	NM902342	140	10NE	8	6·43	33,	−56	250	4	3·82	34,	−49	16·5	23·2
8	NM910344	90	10N	8	4·87	44,	−51	800	3	2·88	38,	−39	16·6	31·2
9	NM857234	10	15W	6	5·95	47,	−54	300	5	4·97	44,	−54	122·5	6·9
10	NM854233	10	15W	6	5·49	34,	−41	800	5	4·52	29,	−40	8·4	27·9
11	NM854235	10	15W	6	5·94	56,	−52	300	6	5·97	56,	−52	155·3	5·3
12	NM836227	350	10W	5	3·40	51,	−22	300	6	5·84	36,	−63	32·0	12·0
13	NM836227	350	10W	5	4·98	49,	−54	300	5	4·98	49,	−56	162·8	6·0
14	NM826218	30	10NW	6	5·99	44,	−46	300	5	4·98	47,	−50	214·0	5·2
32	NM808213	30	10NW	6	5·67	50,	−62	250	5	4·65	51,	−64	11·2	23·7

Note (1.) Statistics in this paper are calculated by the method of Fisher (1953). Notation D = Declination °E of TN; I = Inclination, positive downward; R = Resultant of N unit vectors; k = precision estimate; α_{95} = semi-angle of cone of 95 per cent confidence about the estimated mean direction (D, I). In the present study N is the number of samples, one specimen per core sample.

Table 2

In situ *statistics at other sites*

			Total NRM			Some AF cleaned							
Site	National Grid Reference	No. of samples	R	D,	I	Peak AF (oe)	No. of samples	No. cleaned	R	D,	I	k	α_{95}
15	NM901205	6	5·80	30,	+15	800	6	6	5·81	35,	−05	26·5	13·2
16	NM865202	6	5·66	358,	+64	800	6	6	4·49	46,	−18	3·3	44·0
17	NM858203	5	4·77	4,	−28	250	5	2	4·94	7,	−37	68·5	9·3
18	NM925342	7	6·92	36,	−38	—	7	0	6·92	36,	−38	76·7	6·9
19	NM955333	7	6·92	36,	−47	—	7	0	6·92	36,	−47	74·3	7·0
20	NM983322	7	5·12	32,	−25	250	6	6	4·93	21,	−43	46·9	34·6
21	NN005309	5	4·97	31,	−58	—	5	0	4·97	31,	−58	118·9	7·0
22	NN008305	6	4·73	21,	−21	250	5	5	4·96	15,	−61	89·8	8·1
23	NN013293	5	4·82	55,	−20	—	5	0	4·82	55,	−20	22·6	16·4
24	NN033262	6	5·45	170,	+67	250	6	4	4·93	105,	+72	4·7	34·7
25	NN020273	6	5·69	14,	+51	250	5	5	2·87	Not significant			
26	NN015282	5	4·92	39,	−39	250	5	2	4·51	30,	−29	8·1	28·6
27	NM844152	6	4·75	269,	+58	250	5	5	2·48	Not significant			
28	NM844155	7	6·36	24,	−46	250	7	1	6·63	24,	−44	16·4	15·4
29	NM854165	6	5·98	357,	−37	—	6	0	5·98	357,	−37	28·4	4·0
30	NM856185	6	5·78	0,	−65	250	6	1	5·81	359,	−65	26·8	13·2
31	NM747173	6	4·24	17,	−56	800	6	6	5·52	55,	−73	10·5	21·7
9C	NM857234	3	2·83	148,	+26	Tertiary dyke at site 9							
9B	NM857234	3	2·97	55,	−85	Baked lava at site 9							

The best estimate of the direction of stable NRM is obtained from the 15 localities where dip is known (Table 1); the optimum field for magnetic cleaning at these localities was chosen from criteria of minimum direction change and maximum stability index (Briden 1972). They yield a mean $D = 40°$, $I = -49°$ with precision estimate $k = 43$ after allowance for tilt (Fig. 1(c), Table 3). This corresponds to a palaeomagnetic pole at 2° S, 141° E ($d\psi = 5°$, $d\chi = 8°$). There is a variety of evidence to support this result as a good estimate of the mean geomagnetic field during the igneous episode when the lavas were erupted.

1. Within-site precision is improved by magnetic cleaning at most sites, and in no case does cleaning decrease precision significantly (Table 1).

2. Application of tilt corrections to these 15 site mean directions after magnetic cleaning (Fig. 1(c)) almost doubles the precision of the grouping (Table 3). This constitutes a positive ' fold test ' on the stable remanence with about 92 per cent significance using F-ratio criteria (McElhinny 1964). The tilting itself is probably of Caledonian—most likely Lower Devonian—age and defines a minimum age of the stable NRM. Coupled with the evidence of consistency and stability, it implies that this NRM is primary or nearly so. The combined mean directions at these 15 sites before and after magnetic cleaning, both before and after tilt correction, are all statistically indistinguishable. Cleaning and tilt correction merely improve precision, suggesting that any ' soft ' components removed by cleaning were randomly oriented, and that there is no substantial regional dip.

3. The sites without tectonic control are indistinguishable from those where dip is known in terms of both AF demagnetization characteristics (Fig. 2(b) and (c)) and *in situ* remanence directions before and after AF cleaning (Fig. 1(a) and (b), Table 3), except for sites 25 and 27 which are unstable, and site 24 which is anomalous. Consequently it is unlikely that systematic cleaning and accurate tilt correction at these ' uncontrolled ' sites would alter the best estimate already given. Incorporation of the stable directions from the selected cleaned specimens into the statistics for these sites (Tables 2 and 3) generally improves within-site precision but does not alter their *in situ* overall mean.

4. The Tertiary dyke had scattered NRM, but AF demagnetization showed a small stable component along the typical reversed Tertiary field direction in addition to the large unstable component which dominates the NRM (Fig. 2(b)). The lava adjacent to this dyke was magnetized in an intermediate direction between the reversed Tertiary direction and the typical stable NRM direction in the lavas, which it approached during progressive demagnetization. Lava specimens further from the dyke showed this typical remanence (Fig. 3).

5. The opaque minerals in these rocks range from possibly primary unoxidized titanomagnetite in a few samples, to highly oxidized ilmenohaematites and limonite and leucoxene alteration products. The oxidation state, as judged by reddening at outcrop, varies with position in the flows and also from flow to flow. Direction of magnetization is generally independent of oxidation state, though stability in alternating fields increases with oxidation (Fig. 2). The condition of the opaques is consistent with a combination of primary and deuteric oxidation during the accumulation of the lava pile but the remanence carried by these complex phases may be of chemical rather than thermoremanent origin.

6. Hysteresis of IRM in direct fields up to 15 k oe shows a range of properties. Low coercivity and near saturation in less than 1 k oe may be indicative of the predominance of titanomagnetite. Highly oxidized specimens do not approach saturation in 15 k oe and the dominance of haematite is indicated in such rocks. In all cases NRM is much more stable to AF demagnetization than IRM in the same specimen (Fig. 2(a)). NRM has comparable or greater AF stability than TRM imposed by cooling from 800 °C in a known field, even in unoxidized rocks heated in

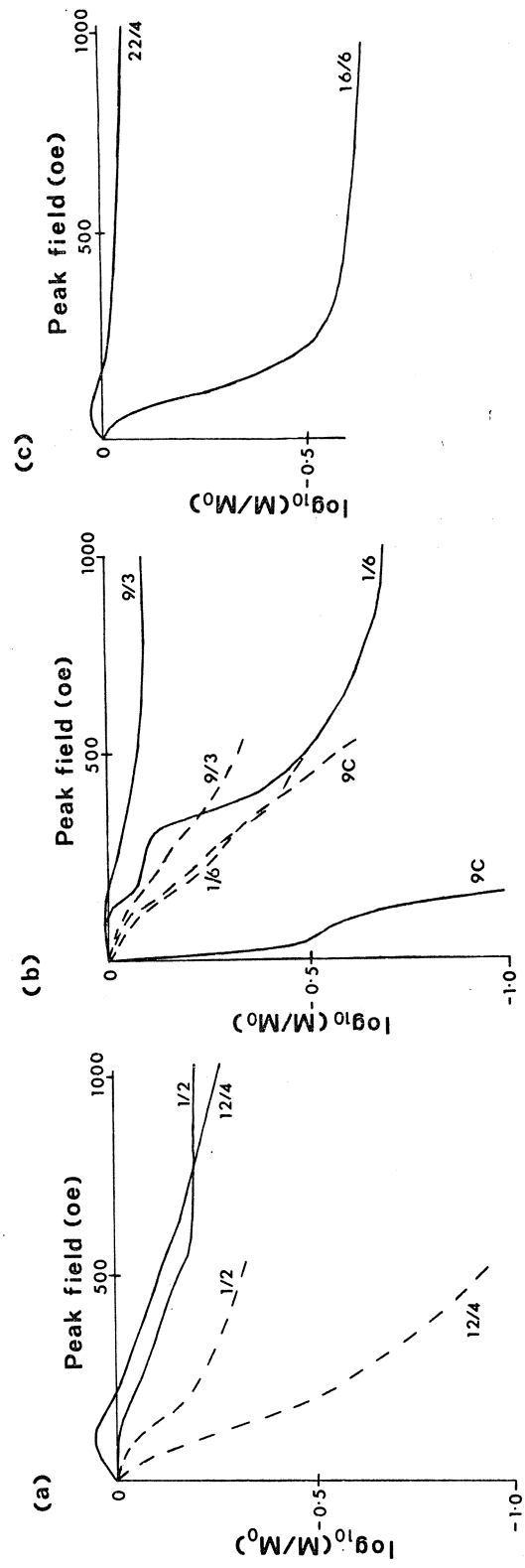


FIG. 2. Alternating field demagnetization curves on semi-log scale. Numbers on the curves denote site/specimen. (a) NRM (solid) and IRM (dashed) in two specimens. (b) NRM (solid) and TRM (dashed) in two extreme specimens from tectonically controlled sites and Tertiary dyke 9C. (c) NRM in two extreme specimens from tectonically uncontrolled sites.

Table 3

Overall statistics and comparisons with previous work

	Total NRM				AF cleaned			
	k	α_{95}	D, I *in situ*	D, I tilt corrected	k	α_{95}	D, I *in situ*	D, I tilt corrected
15 sites with tectonic control (this study, Table 1)	23·8	7·9	39, −46(1)		24·3	7·9	38, −50	
	27·0	7·4		41, −44	**42·7**	**5·9**		**40, −49**
14 sites without tectonic control (this study, Table 2, sites 24, 25, 27 excluded)	6	18	25, −32		12	12	28, −42	
5 sites of Embleton (1968), assuming uniform tilt of 15° to SE	16	20	69, −14(2)	64, −20(2)	23	17	72, −19(2)	66, −24(3)
Sites in common with Embleton (1968)								
Site 1, this study 5 specimens	132	7	47, −25	37, −29	142	6	52, −30	40, −35
Embleton site L15 10 „		29	67, −3(2)	65, −8		23	66, −6(2)	64, −11
Site 3, this study 7 „	4	33	52, −25	45, −27	155	6	53, −34	43, −26
Embleton site L20 28 „ (25 AF cleaned)		1	73, −48(2)	56, −53		1	73, −48(2)	56, −53
Site 6, this study 7 „	92	6	42, −40	28, −35	33	11	44, −44	27, −39
Embleton site L18 10 „		17	70, −20(2)	65, −25		17	70, −20(2)	65, −25

Notes

(1) Strictly, site 12 should have been omitted from this Total NRM analysis because its mean direction just fails the test of significance at the 95 per cent probability level (Table 1); it passes the test after AF cleaning, and the procedure adopted here facilitates direct comparison of total NRM and AF cleaned statistics at the 15 tectonically controlled sites without serious loss of rigour.

(2) Calculation by the present authors from the published data, assuming that a tilt correction 15° to the SE had been applied to the published data.

(3) Calculation by Briden *et al.* (1973) from the published data.

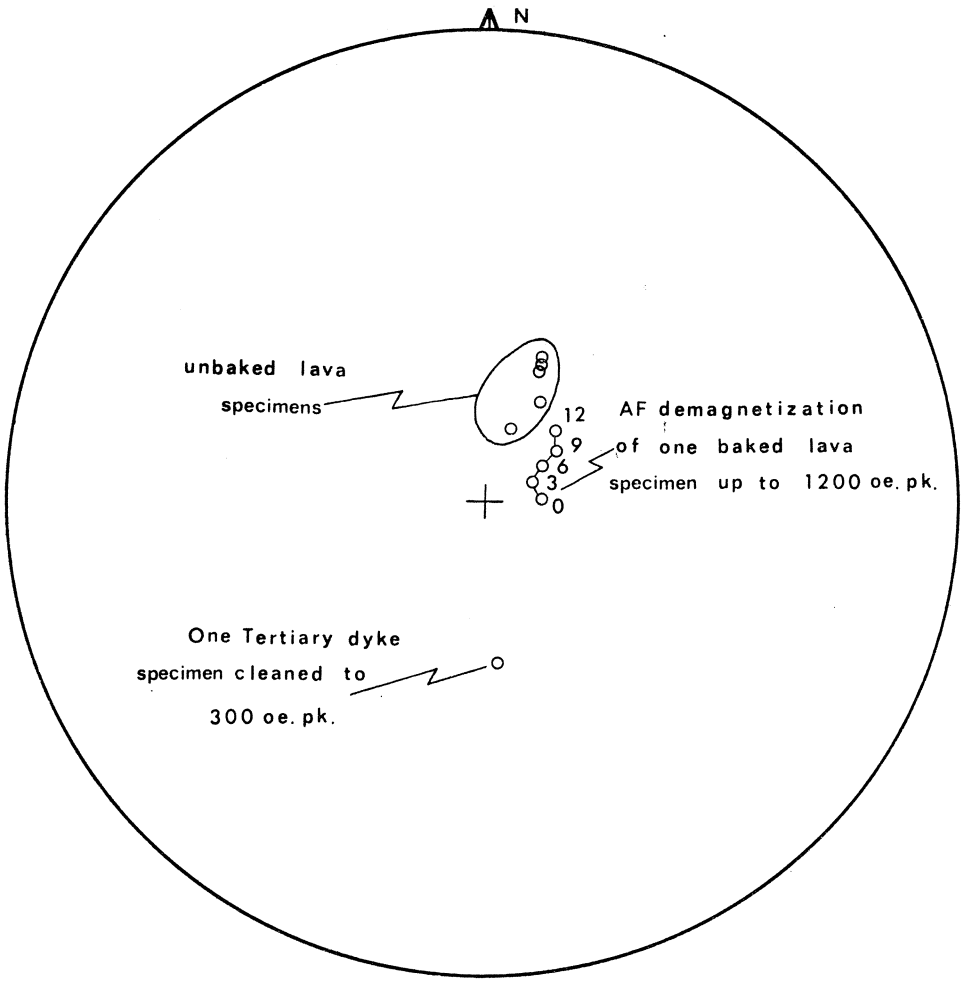


FIG. 3. Stereographic projection showing coplanar distribution of *in situ* AF cleaned remanence in Tertiary dyke and unbaked lava at site 9, with progressive demagnetization directions in a baked lava specimen at that site. Numbers denote peak alternating field in hundreds of oersteds. All points are on the upper hemisphere.

air (Fig. 2(b)). The opposite was found only in the Tertiary dyke in which NRM was known to be dominated by a soft component (curves 9C in Fig. 2(b)).

Regional implications

Embleton's (1968) study of Scottish lavas included five sites in the area of the present work, at which AF cleaned remanence directions were corrected for a general regional tilt. The new results presented here differ from those of Embleton (1968) by typically 20° in declination and inclination of site mean directions at the three sites common to the two studies, and also in the overall mean directions (Table 3). The differences are not explained by the different tilt correction procedures which were adopted. If small-sample bias were the cause Embleton's results might be preferred because he measured more specimens at each site; on the other hand the new result could be favoured because many more sites were involved. On balance the new result

is the more reliable because it is based on more precise tectonic tilt corrections leading to a positive fold-test, and is supported by the additional evidence given in the previous section.

The new data are in accord with data from rocks of similar age in England and the mainland of Scotland in yielding a Siluro–Devonian palaeomagnetic pole near 0°, 140° E. This is not to say that all problems related to this topic have now been solved. The Foyers Granite is commonly believed to be of this age, yet it yields a significantly different palaeomagnetic pole (Kneen 1973). There is little geological control on its age and one explanation would be that it is actually older, and some isotopic data tend to support this view (S. R. Charlton 1974, private communication). Also the well-documented discrepancy between British and Norwegian data remains together with the uncertainties in interpretation of results from Orkney and Shetland, but these have been adequately discussed elsewhere (Briden, Morris & Piper 1973; Storetvedt 1970; Storetvedt & Petersen 1972).

The Lorne Plateau volcanics have long been regarded as of similar age to the extensive Lower Old Red Sandstone age volcanics in the Midland Valley of Scotland, a correlation supported by the published K–Ar age data. The slightly older isotopic ages now reported from Lorne, together with the contrast between the single magnetic polarity of the Lorne sequence and the several reversals in the Midland Valley (Sallomy & Piper 1973) now suggest that this correlation may not be precise. The oldest horizons in the Midland Valley which are known to have opposite polarity to the Lorne lavas occur in the upper part of the Crawton Group, and probably define the minimum stratigraphic age of the Lorne sequence.

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